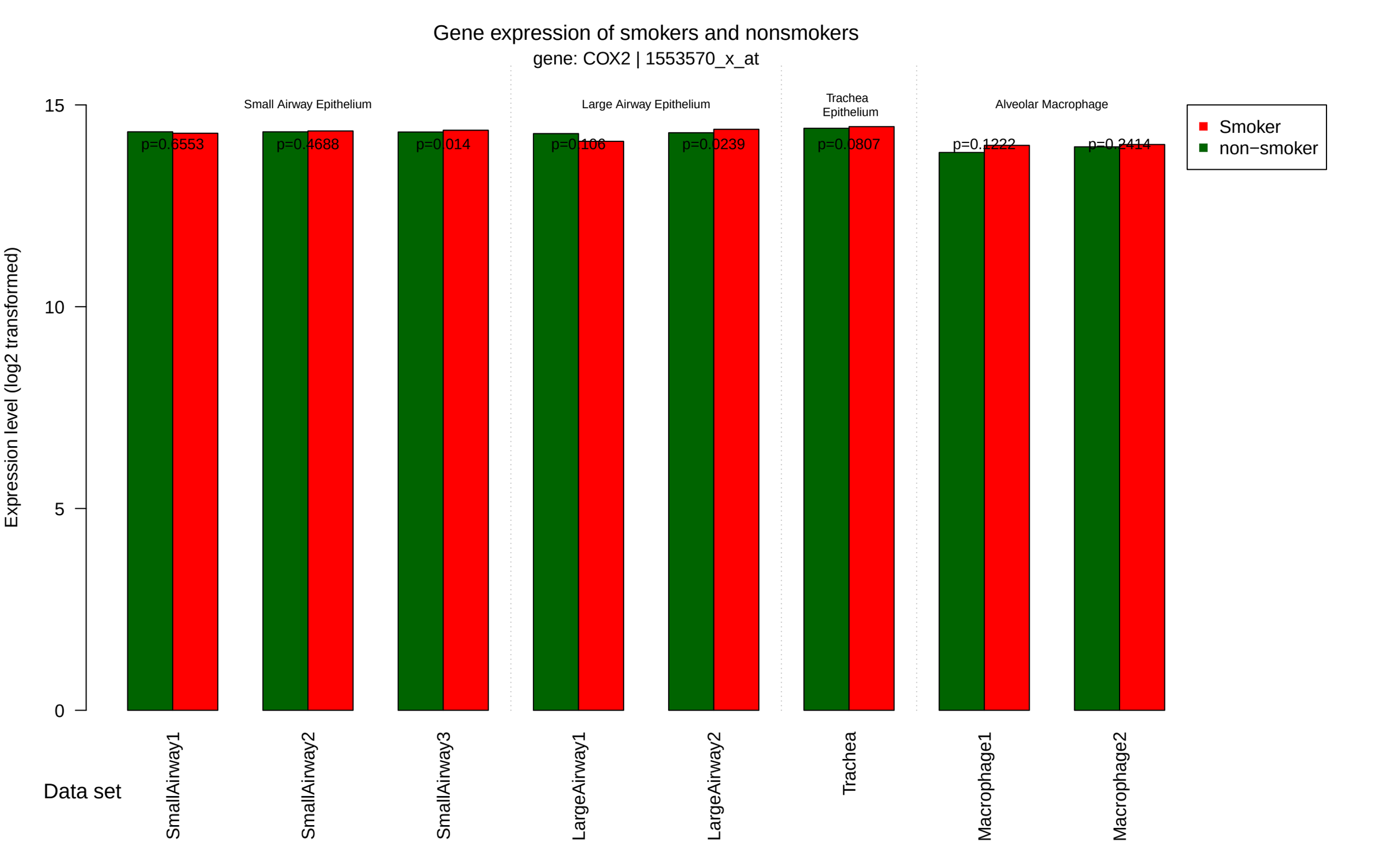
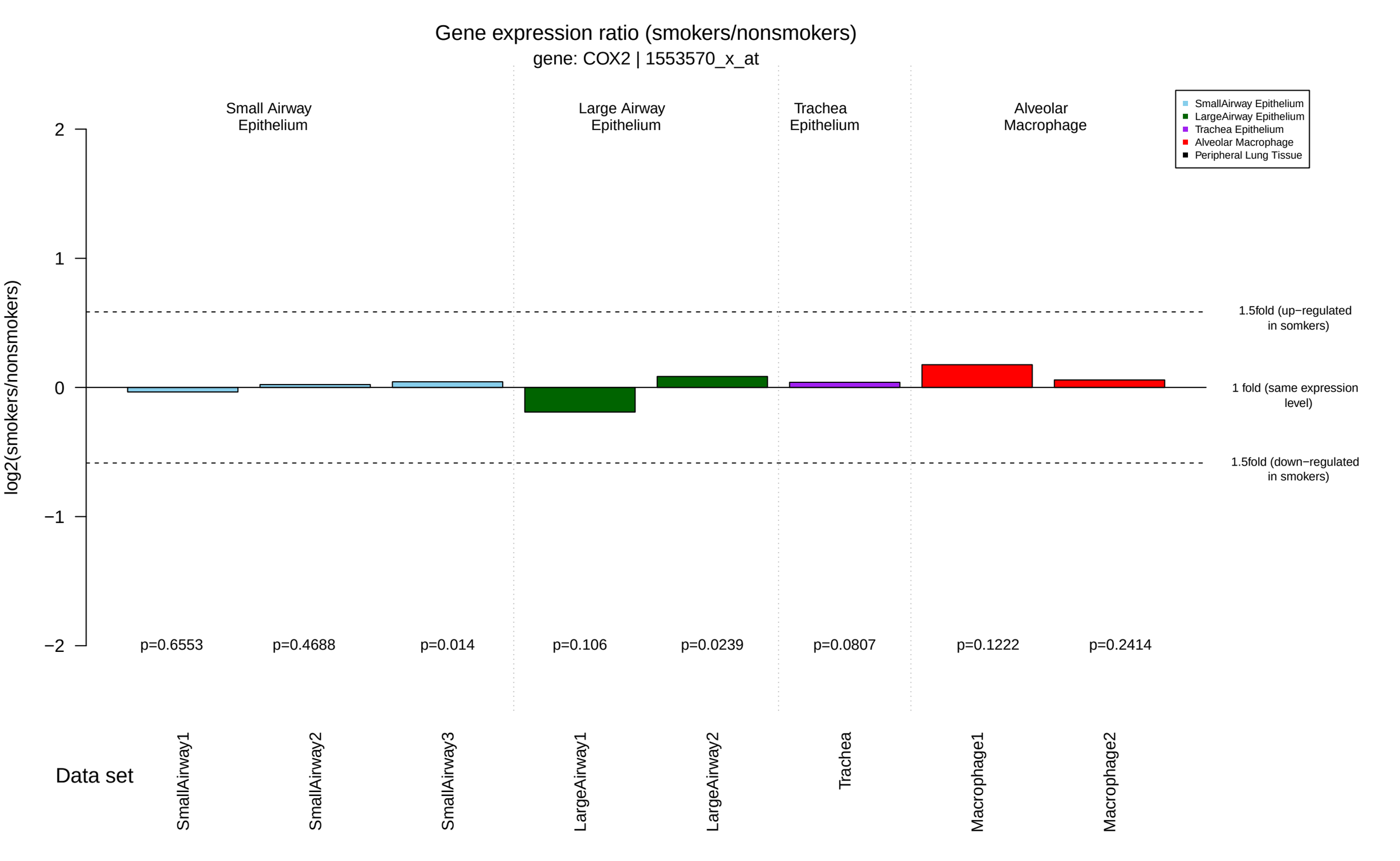
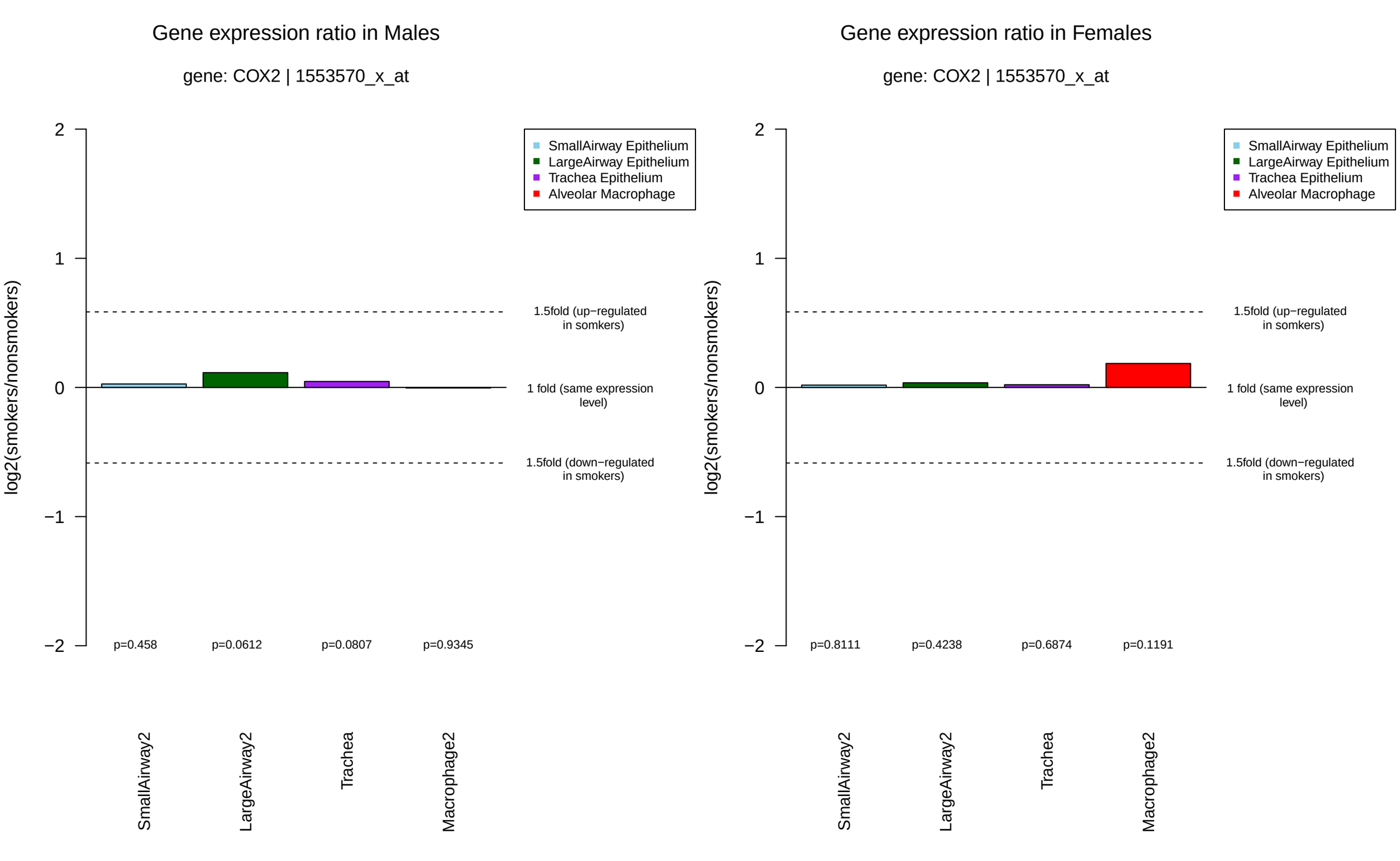
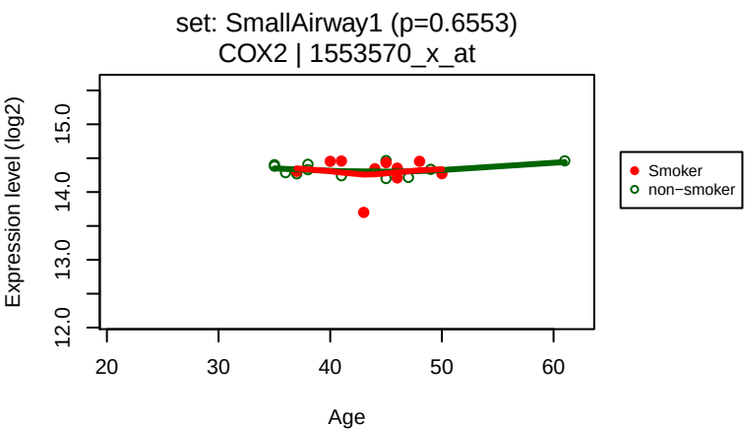
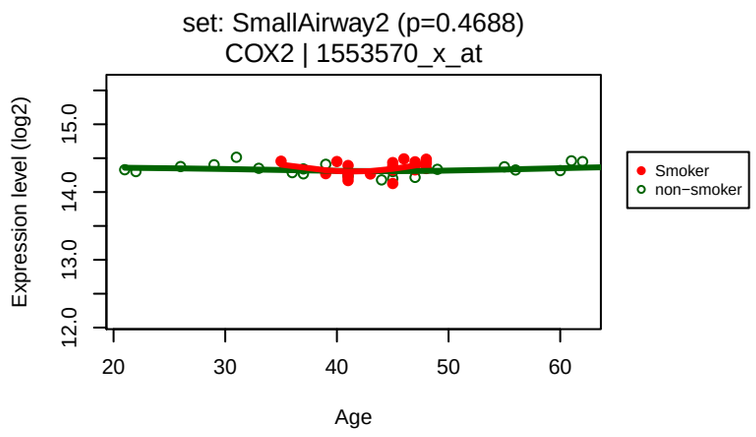
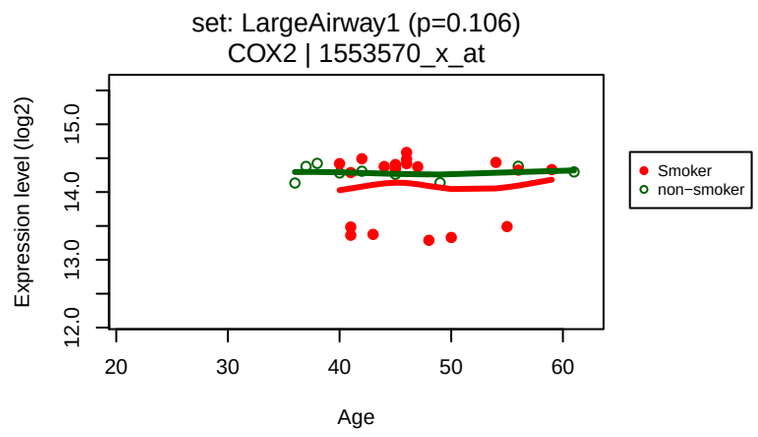

Gene expression of smokers and nonsmokers
gene: COX2 | 1553570_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6553
p=0.4688
p=0.014
p=0.106
p=0.0239
p=0.0807
p=0.1222
p=0.2414
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

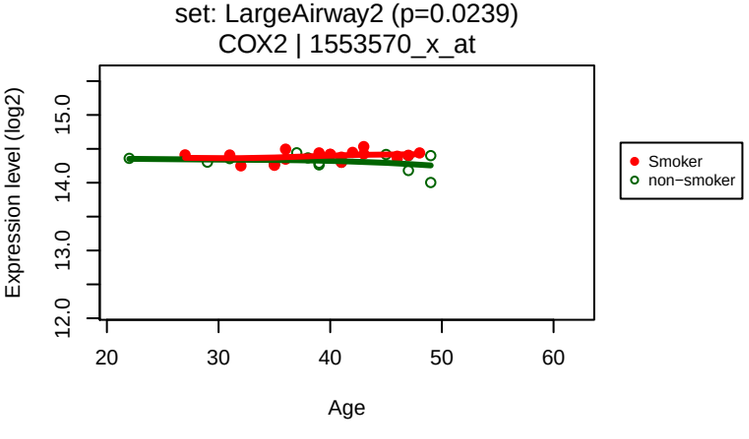

Gene expression ratio (smokers/nonsmokers)
gene: COX2 | 1553570_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6553
p=0.4688
p=0.014
p=0.106
p=0.0239
p=0.0807
p=0.1222
p=0.2414
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

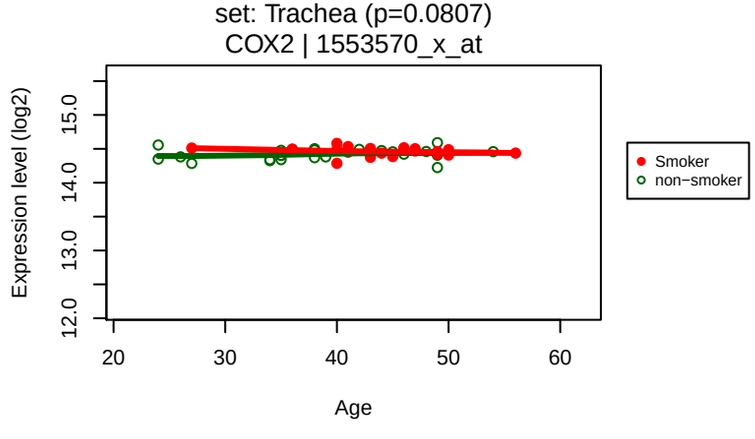

Gene expression ratio in Males
gene: COX2 | 1553570_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.458
p=0.0612
p=0.0807
p=0.9345
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COX2 | 1553570_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8111
p=0.4238
p=0.6874
p=0.1191
SmallAirway2
LargeAirway2
Trachea
Macrophage2

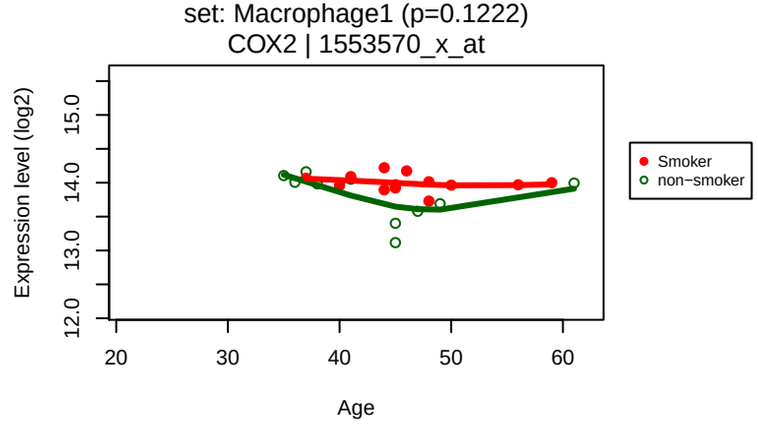

set: SmallAirway1 (p=0.6553)
COX2 | 1553570_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

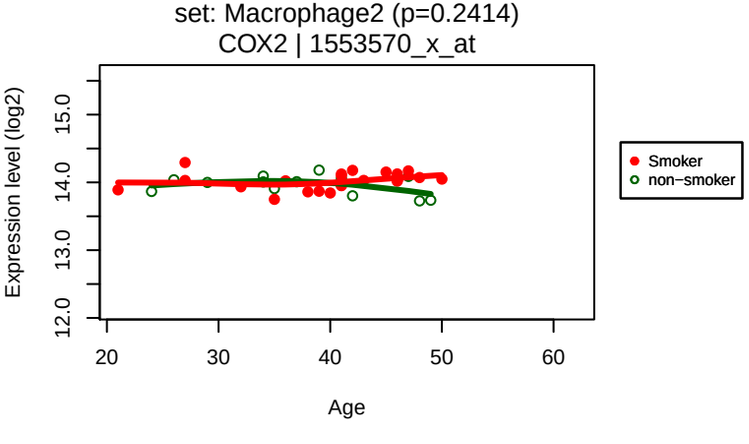

set: SmallAirway2 (p=0.4688)
COX2 | 1553570_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker

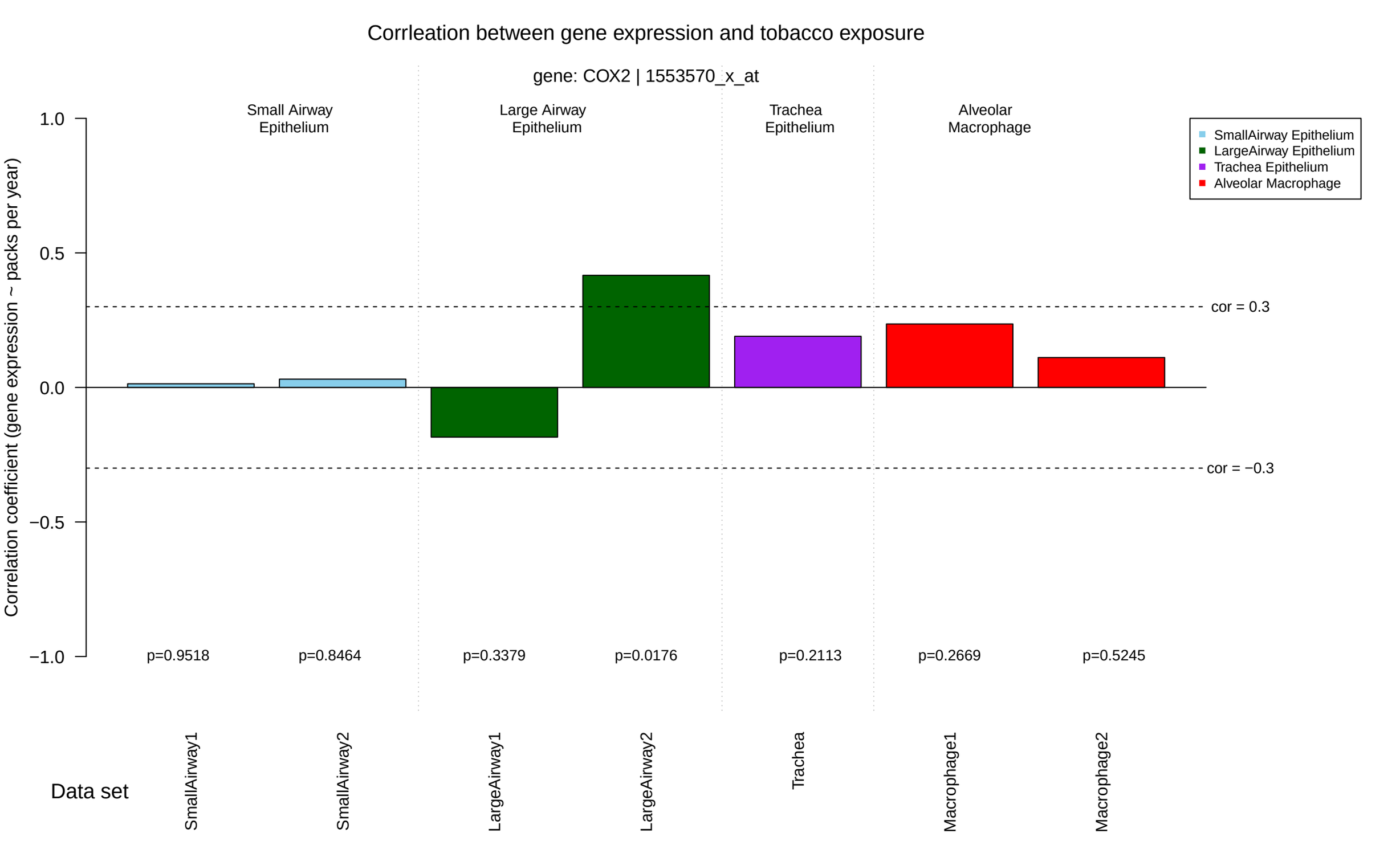

set: LargeAirway1 (p=0.106)
COX2 | 1553570_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0239)
COX2 | 1553570_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.0807)
COX2 | 1553570_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1222)
COX2 | 1553570_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2414)
COX2 | 1553570_x_at
Expression level (log2)
12.0
13.0
14.0
15.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COX2 | 1553570_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9518
p=0.8464
p=0.3379
p=0.0176
p=0.2113
p=0.2669
p=0.5245
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2